Mem²Evolve: Towards Self-Evolving Agents via Co-Evolutionary Capability Expansion and Experience Distillation

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HarnessRSI 65 ★★

evidence: retained

Paper

1 What changes, what is kept

Asset Memory
Tools + expert-agent specs
roles / expertise / suggestions

Experience Memory
Tool implementation recipes
Agent strategies + failure fixes

1 Plan, retrieve, reuse first
Top expert + relevant tools above threshold; missing capability → create on demand.

2 Experience guides new capabilities
Use remembered recipes + web documents to create tools and expert-agent specs.

3 Execute → LLM judge
Task + trajectory + answer → success / critique.
No ground-truth labels for backward evolution.

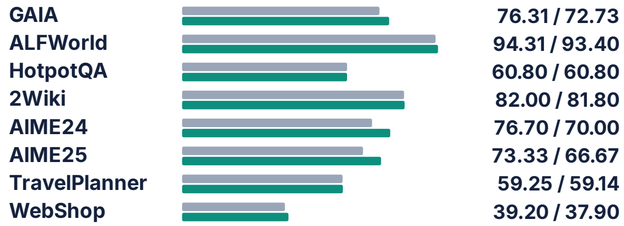
4 Test and repair before admission
Task succeeds AND all synthesized tests pass; else revise + regenerate tests until pass.

5 Keep assets; append lessons
Store validated assets and success/failure lessons.
Both memories guide the next task.

Two memories co-evolve; no model-weight update
No stated retry cap or independent regression gate

2 Reported average: 70.24 vs 63.78

GPT-5-chat · eight benchmarks · vs Alita (Table 2)



Mixed benchmark metrics, not pooled Pass@1. No SD/CI or run count. GAIA Total aggregation is unresolved across rows; reported totals retained.

4 Where the gain is thin

- **HotpotQA ties:** 60.80 vs AFLOW 60.80. **2Wiki:** 82.00 vs Direct 81.80. **TravelPlanner:** 59.25 vs SwarmAgentic 59.14; uncertainty unreported.
- **Tool creation matters mainly on GAIA/math.** Removing it: GAIA 76.31 → 21.69; AIME25 73.33 → 60.00. ALFWorld barely shifts: 94.31 → 94.10; WebShop 39.20 → 39.05 (Table 6).
- **Tool memory adds little here:** full vs removal, ALFWorld 94.31 vs 94.25; WebShop 39.20 vs 39.15. One backbone only.
- **Leakage unresolved:** task context/outcomes retained; subset relatives and web contamination not audited.

3 Controls and missing controls

Ablation averages · Tables 3 / 6

Full system	70.24
No tool creation	59.96
No expert creation	68.52
No tool memory	67.11
No agent memory	65.51

No reported variance. Agent-memory removal: TravelPlanner 59.25 → 49.36, ALFWorld 94.31 → 88.06.

Experience-guided creation · Table 4
Mean first-pass validity **53.1% → 72.4%**; mean repair iterations **1.01 → 0.48**. No uncertainty or generation counts; not a total-cost measure.

Retained memory · §5.3–5.4
Same-task: accumulate on 25%‡ / 50%‡, test remainder. Transfer: GAIA memory → seven other benchmarks, gains over empty initialization. Freeze during evaluation and near-duplicate separation unspecified.

Fairness and resource accounting
Alita has an Experience Pool (OpenAlita implementation). No same dual-memory/test gate granted to baselines in a controlled comparison; no matched-compute row.

Unreported: default-selection split, baseline tuning budgets, token/$/time or memory-growth cost.

Takeaway

Experience helps create reusable capabilities, and those capabilities feed new experience. Ablations support the coupling, but much of the tool-creation gain comes from GAIA and math; budget-controlled self-improvement is not established.

Our read · HarnessRSI

Score 65 · ★★ · evidence: retained
Reports: partial held-out evidence, transfer, ablations, tool-validity proxy.
Unclear: variance, matched budget, optimizer cost, leakage guards, improver comparison.